DDD-W-80E
June 29, 1987
SUPERSEDING
DDD-W-80D
October 12, 1976

FEDERAL SPECIFICATION

WASHCLOTH, TERRY, COTTON

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of cotton washcloths.

1.2 Classification.

1.2.1 Types. The cotton washcloth shall be of the following types as specified (see 6.2)

Type I - 16 ounce per dozen, minimum

Type II - 11 ounce per dozen, minimum

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents in effect on date of invitation for bids or solicitation for offers form a part of this specification to the extent specified herein.

Federal Specifications

A-A-203	- Paper, Kraft, Untreated
V-T-276	- Thread, Cotton
DDD-L-20	- Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S Army Natick Research Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7210

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PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, Sealing and Securing

Federal Standards:

FED-STD-123 - Marking for Shipment (Civil Agencies)
FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams, and Stitchings

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements, as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA, New York, NY, Philadelphia, PA, Washington, DC, Atlanta, GA, Chicago, IL, Kansas City, MO, Fort Worth, TX, Denver, CO, San Francisco, CA, Los Angeles, CA, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items, Clothing, Personal Equipment and Equipage, General Specification For

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Federal Regulations

Federal Trade Commission

Rules and Regulations Under the Textile Fiber Products Identification Act

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3 REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished washcloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarns. The yarns shall be single or two ply yarns as specified in table I. The pile warp yarns shall be 100 percent cotton. The ground warp and filling yarns shall be either cotton or polyester/cotton which may contain up to 70 percent polyester with the remaining percentage cotton.

3.3 Physical requirements. The finished washcloths shall conform to the requirements specified in table I when tested as specified in 4.2.5

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TABLE I. Physical requirements

Type	Weight 1/ per dozen (ounces), minimum	Yarns per inch, minimum		Ply warp		Filling Ply	Finished dimensions (inches)		Breaking strength (pounds), minimum	
		Warp	Filling	Ground	Pile		Each	Tolerance	Warp	Filling
I	16	68 2/	35 2/	2 or 1	2 or 1	1	12 by 12	± 1/2	40	35
II	11	59	29	2 or 1	2 or 1	1	12 by 12	± 1/2	25	25

1/ The washcloths shall reach equilibrium under Standard Conditions as defined in FED-STD-191 prior to determination of weight.

2/ Either of these values may fall below this minimum by not more than three yarns, provided the combined warp and filling yarn total does not fall below 100 yarns per inch.

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3.3.1 Weave.

3.3.1.1 Type I. The type I washcloth shall consist of a ground warp, pile warp, and filling yarns woven as a single-loop terry weave exclusive of a border. As alternates, the type I washcloth shall have a plain woven border 1/4 to 1/2 inch wide either on both sides and both ends or on both ends only. The border shall be located not more than 1 inch from the ends and sides or from the ends only, whichever is applicable

3.3.1.2 Type II. The type II washcloth shall consist of a ground warp, pile warp, and filling yarns woven as single-loop terry weave. The border shall be on both sides and both ends, or on both ends only. The border measurement, from the edge towards the body, shall be in this order 3/8 to 5/8 inch plain weave, 1/8 to 3/8 inch terry, 1/16 to 3/16 inch plain weave, 1/8 to 3/8 inch terry; 1/16 to 3/16 inch plain weave, terry body.

3.3.2 Thread. The thread for stitching the type I washcloth shall be cotton, ticket No. 30, 3 ply, type IA2 of V-T-276. The thread for stitching the type II washcloth shall be cotton, ticket No. 50, 3 ply, type IA2 of V-T-276.

3.4 Construction.

3.4.1 Edge finishing. All four edges shall be overedged stitched with 10 to 14 stitches per inch of stitch type 504 or 505 and stitching type EFd-1 as specified in FED-STD-751. All ends of stitching, thread breaks, skipped and runoff stitches shall be overstitched 3/4 to 1 inch There shall be no loose or untrimmed threads on the finisned washcloths.

3.5 Color. The washcloths shall be desized and bleached white to match the standard sample when available. The bleached washcloths shall show no more yellowing or discoloration than that shown by the standard sample after 20 hours exposure to light (see 4.2.5). When no standard sample is available, the discoloration shall be no more than the rating of 'fair' when tested as specified in 4.2.5.

3.5.1 Matching. The bleached washcloths shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles, and shall be a good approximation to the standard sample under incandescent lamplight at 2300 ± 200K.

3.6 Nonfibrous material content The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 5.0 percent when tested as specified in 4.2.5.

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3.7 Marking for identification. When marking for identification is specified (see 6.2), each washcloth shall have an identification label conforming to the requirements of type VI, class 4 of DDD-L-20, except that all references to size shall be deleted. The right side of the label shall be caught in the overedge stitching and the bottom edge of the label shall be 1/2 to 3/4 inch from the lower corner of the washcloth. The end item is intended to be laundered.

3.7.1 Fiber identification. Each washcloth shall be labeled for fiber content in accordance with the Textile Fiber Products Identification Act.

3.8 Workmanship. The finished washcloths shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection In accordance with 4.1 components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded amended, modified, or qualified in this specification or applicable purchase specification

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4.2.2 End item visual examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of washcloths. The sample unit shall be one washcloth. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

Defects
Cut, hole, tear, or smash
Three or more contiguous broken or missing yarns (clearly visible)
Terry loops improperly formed over an area of 1 square inch or bald areas of 1/2 square inch or more
Spot, stain, streak, clearly visible
Any trash, motes, leaf, or seed coat fragments
Objectionable odor
Color or shade not as specified
Wrong stitch type or stitches per inch not as specified
Any overedge stitching missing, three or more consecutively skipped stitches, run-off stitching, loose stitch tension resulting in a loosely secured edge
Identification label missing, incorrect, illegible, insecurely attached or misplaced (when applicable)
Border not as specified (when applicable)
Any loose or untrimmed threads
Any thread breaks, runoff stitching, and ends of stitching not overstitched 3/4 to 1 inch

4.2.3 End item dimensional examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of washcloths. The sample unit shall be one washcloth. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Defect
Overall length or width less than 11 1/2 inches
Overall length or width more than 12 1/2 inches

4.2.4 End item weight examination. The end item shall be examined for the weight requirements specified in table I. The lot size shall be expressed in units of washcloths. The sample unit shall be one washcloth. Any unit not meeting the minimum weight requirement shall be considered a defect. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4.2.5 End item testing. The end item shall be tested for the characteristics specified in table II. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The lot size shall be expressed in units of washcloths. The sample unit shall be five washcloths. The sample size (number of sample units) shall be as shown below. The lot shall be unacceptable if one or more sample units fail to meet any test requirement.

Lot size	Sample size (sample unit)
800 or less	2
801 up to and including 22,000	3
22,001 and up	5

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Yarns per inch	3.3	5050
Breaking strength	3.3	5100
Weave	3.3.1	Visual 1/
Discoloration	3.5	5660 1/ 2/
Nonfibrous material	3.6	2611

1/ One determination shall be made on each sample unit and the results reported as "pass" or "fail".

2/ Method 5660 shall be used, except that exposure of the specimen and standard sample shall be for 20 standard facing hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "pass". Any specimen showing discoloration greater than that of the standard sample shall be rated "fail". When no standard sample is available, the discoloration of the specimen shall be considered satisfactory if it is not worse than the rating of "fair" indicated in Method 5660.

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4.2.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of bundles or bags per container is more or less than specified Number of washcloths per bag or bundle is more or less than specified 1/

1/ For this defect, three bundles or bags shall be examined from each shipping container in the sample.

4.2.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified

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Examine	Defect
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect; illegible, of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A Twelve washcloths of one type only shall be evenly stacked, flat or center folded, and then completely wrapped in 30-pound minimum basis weight kraft paper conforming to A-A-203. Each paper wrapped bundle shall be cross-tied with cotton tape or twine, or secured with gummed paper tape conforming to type III, grade B of PPP-T-45. Alternatively, the stack of twelve washcloths shall be unit packed in a close-fitting clear polyethylene film bag of 0.00125 inch thickness (± 25 percent tolerance). The polyethylene bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and to the formed edges of the bag. The bag closure shall be effected by heat sealing with the heat seal made as close as possible to the open end or by means of a mechanical tie (paper or vinyl plastic covered soft steel wire, aluminum band, etc.). A 1/4 inch maximum diameter hole shall be made at one corner of each polyethylene bag to allow the excess air to escape The bag may be fabricated from polyethylene film tubing or sheeting.

5.1.2 Commercial. Washcloths shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Three hundred washcloths of one type only, preserved as specified in 5.1, shall be compactly packed within a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636 The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636 Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.2 b. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2) Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.2.2 Level B packing. Three hundred washcloths of one type only, preserved as specified in 5.1, shall be compactly packed within a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each container shall be fitted with a taped box liner conforming to class domestic, variety SW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.2.6.

5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.2.6.

5.2.3 Commercial packing. Washcloths, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), washcloths, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be in accordance with appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. Marking shall be in accordance with 5.4.1 or 5.4.2, as specified (see 6.2).

5.4.1 Civil agencies. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with FED-STD-123.

5.4.2 Military requirements. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.3 Polyethylene bagged packages. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

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6. NOTES

6.1 Intended use. The washcloths are intended for use by the Federal agencies and the Department of Defense.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number, and date of this specification.
b. Type of washcloth required (see 1.2).
c. When marking identification is required (see 3.7).
d. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).
h. Marking required (see 5.4).

6.3 Standard sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 Subject term (key word) listing.

Terry
Washcloth

MILITARY INTERESTS

Custodians.

Army - GL
Navy - NU
Air Force - 99

Review Activities

Army - MD
DLA - CT

User Activity

Air Force - 45

CIVIL AGENCY COORDINATING ACTIVITY

GSA - FSS

PREPARING ACTIVITY.

Army - GL

Project No. 7210-0276

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
DDD-W-80E	Washcloth, Terry, Cotton

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street City State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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